

Secure Transmission Technology for Aviation Broadband Communication System

Zhijun Wu*, Minghui Sun†, Tao Fang†, Jin Lei*, Meng Yue*

*College of The Safety Science and Engineering, Civil Aviation University of China, Tianjin, China

†College of Electronic Information and Automation, Civil Aviation University of China, Tianjin, China

Emails: {zjwu, 2022021047, 2023021003, jlei, myue}@cauc.edu.cn

Abstract—As air transportation expands and present civil aviation communications systems reach their capacity limits in high-density locations, the requirement for new aeronautical communications technology becomes becoming increasingly important. Aviation broadband communication system (ABCS), as a key infrastructure for building the next generation air traffic control system, possess advantages such as low latency, high reliability, and large bandwidth to meet the growing demands of air transportation. However, these systems lack effective information security measures which pose security risks to ground-to-air data transmission. This paper proposes a secure transmission technology for ABCS based on the SM series cipher. The proposed approach utilizes SM4 for data encryption, SM3 for data integrity protection, and Timed Efficient Stream Loss-tolerant Authentication (TESLA) for message source authentication. The experimental results show that this scheme has faster encryption and decryption speeds, lower security overheads, and it can effectively resist the tampering attack, which meets the security requirements of ABCS.

Keywords— ABCS, SM4, SM3, TESLA, secure transmission

I. INTRODUCTION

Today's air-to-ground communication system mainly relies on VHF, which has been rapidly developed and applied on a large scale. However, the frequency band of VHF is limited and is facing the influence of gradual saturation in high-density areas [1]. The transition from narrowband to broadband aviation communication systems must be encouraged in order to support long-term sustainable air transportation and implement new air traffic management (ATM) services and technologies in the future to satisfy the increasing demand for high reliability, high bandwidth, and high speed of aviation communication services. Strengthening the technical research and development of the new generation of aeronautical communication systems has become the main development trend and indispensable choice for the construction of modern aeronautical communication systems around the world [2].

As early as 2002, NASA and the European Organization for the Safety of Navigation (EUROCONTROL) have begun to jointly organize research work for the next generation of aeronautical communication systems [3]. In 2007, the International Telecommunication Union (ITU) passed a resolution on the deployment of frequency resources for ABCS at the World Radio Conference: L-Band Digital Aeronautical Communications System (LDACS) is deployed in the L-band (890 ~ 1164 MHz)[4]. In order to promote the application of the new generation of ABCS and establish the broadband communication capability of aircraft with aviation safety communication as the core, the Civil Aviation

Administration of China (CAAC) issued the implementation roadmap of China Civil Aviation ATM Modernization Strategy (CAAMS) in March 2020 [5]. The CAAMS implementation roadmap clearly points out that the future aviation communication will show the trend of broadband ground-to-air communication in the whole stage of flight, and will carry out the test and application of route aviation broadband communication technology as one of the medium and long-term goals. Encouraging the use of the latest broadband communication technology in the civil aviation sector will effectively promote the new smart civil aviation technology and offer a solid assurance for the industry's high-quality growth.

ABCS is the key information infrastructure for aviation transportation security. Its data link is in an open communication state. The malicious attack of the attacker on the data link will cause damage to the system, which will affect the flight safety of the aircraft [6]. Therefore, in the research of ABCS, information security will be regarded as a key issue. In the process of data transmission, the data link of ABCS needs security measures such as encryption, integrity protection and verification and source authentication to ensure the confidentiality, integrity and availability of data [7], so that it has the ability to effectively resist network attacks such as data eavesdropping, tampering and replay, so as to achieve end-to-end data security transmission. In order to realize the above security goals, there is an urgent need to research and develop robust information security guarantee technology for data transmission of ABCS, so as to make the safe and effective operation of ABCS, and to promote the safe and reliable development of civil aviation airway navigation and communication and surveillance information transmission.

The airline electronic engineering committee (AEEC) formulated the message security standard of aircraft communication addressing and reporting system (ACARS) and gave the encryption scheme for character coding[8]. For the first time, German Aerospace Center (DLR) proposes a complete LDACS network security architecture and algorithms to support its implementation, and embeds user data encryption and data integrity functions in the existing protocol stack. It proposes the option of using AES-GCM for confidentiality protection, HMAC or AES-GCM for integrity protection, and providing broadcast authentication protocol for ensuring the legitimate source of data[9]. It provides a design basis for the security research of ABCS.

II. KEY ALGORITHMS AND SECURE PROTECTION ANALYSIS

In order to realize the secure data transmission of ABCS, this paper designs the secure transmission technology of ABCS based on the SM series cipher. SM4 is used to encrypt and decrypt the communication data of ABCS, SM3 protects

the integrity of the data, and TESLA is used to realize the authentication of the message source, so as to realize the security protection of the user data as well as the control data of ABCS, and the key algorithms are expressed as follows.

A. SM4 Algorithm

Due to the limited resources of the aircraft station (AS) in the system, the processing capacity is lower than that of the ground equipment, so there is a low overhead communication requirement. The domestic SM4 cipher is a block cipher with fast encryption speed and low computational overhead. It has the advantages of high security, high efficiency, lightweight design, and self-controllability, and is suitable for the security requirements of ABCS. The input of SM4 algorithm is 128 bits plaintext, which has been transformed and iterated many times. The last encrypted output is spliced with the last encrypted data for the next encryption, and finally the ciphertext with a length of 128 bit is output by reverse order transformation [10].

B. SM3 Algorithm

According to the data integrity protection requirements of ABCS, this study uses the domestic hash cipher SM3 algorithm to realize the integrity verification of communication data. The SM3 algorithm adopts the Merkle-Damgard structure. First populate the message m with $m' = padding(m)$, forming a bit string with a total length of 512 bits multiples. The filled message m' is grouped according to 512 bits, and the grouped data is iteratively compressed and extended to obtain the extended message word for hash value. HMAC algorithm is a keyed message authentication code algorithm, which is used to verify data integrity and source legitimacy. The definition of the HMAC algorithm process is shown in equation 1.

$$HMAC(K, text) = H(K \oplus opad, H(K \oplus ipad, text)) \quad (1)$$

Among them, K is the shared key extracted by both parties, H is the Hash function, $text$ is the data to be verified, and $opad$ and $ipad$ are constants defined by the algorithm. The process of SM3-HMAC is shown in Fig. 1 [11]. Due to the high efficiency of the SM3-HMAC algorithm, it does not impose a burden on performance in an ABCS with limited computing resources, while ensuring fast data authentication.

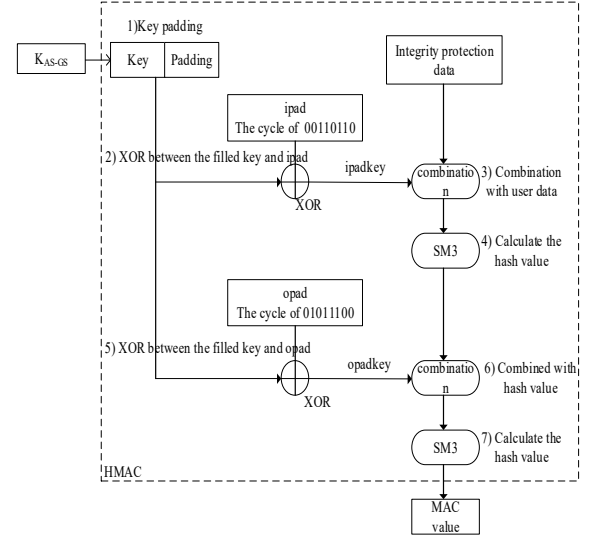


Fig. 1. The specific process of SM3-HMAC

C. TESLA

In order to realize the message source authentication of the ABCS, TESLA broadcast authentication protocol is adopted. The core idea is to use the delayed key to experiment with asymmetric authentication. The working principle is shown in Fig. 2 [12]. The working principle of TESLA is to divide the time into equal intervals T_{int} , and generate a one-way key chain k_i for a specific time interval, and the direction of generating the key chain is opposite to the direction of using the key. The sender uses the key to generate MAC and sends it to the receiver along with the key at the previous interval. Therefore, the receiver can verify the message source authentication after receiving the delayed published key. The calculation amount of this method is smaller than that of the asymmetric method, which can realize the reliable transmission of the message. However, in order to use TESLA correctly, the sender and receiver must have loose time synchronization and must transmit a series of pre-parameters required by TESLA in a secure way.

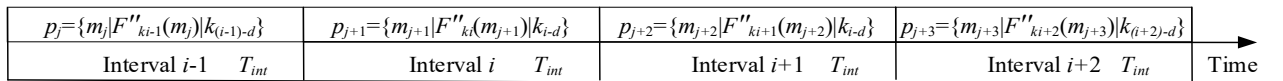


Fig. 2. TESLA protocol flow

D. Channel Protection Analysis of ABCS

ABCS is divided into forward link (FL) and reverse link (RL). For different types of messages, they are transmitted on different channels [13]. Data Channel (DCH): For DCH, the message is transmitted in plain text in the process of air-to-ground communication, and it is not encrypted by any means. This information includes not only the aircraft's flight number, take-off and landing time, current geographical location, but also meteorological information. The loss or tampering of data is likely to cause economic losses to airlines,

and even threaten the lives of passengers and crew on the plane. There are huge security risks. Therefore, it is necessary to encrypt and protect the integrity of user data to realize the secure transmission of user data in aviation broadband system.

III. ABCS SECURITY PROTECTION SCHEME DESIGN

A. ABCS Security Protection System

The security features implemented in ABCS can be realized by adding security layers. By adding a lower layer of security, a security layer independent of higher-level security

functions is provided between the aircraft and the ground station, so that ABCS can communicate safely. The

information security protection architecture of ABCS is shown in Fig. 3 [14].

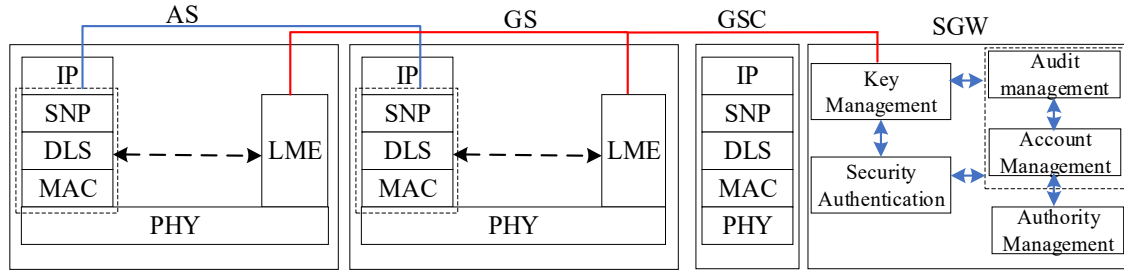


Fig. 3. ABCS Security protection architecture

The entity of ABCS mainly consists of four parts: Aircraft Station (AS), Ground Station (GS), Ground Station Controller (GSC) and Security Gateway (SGW). The SGW mainly integrates key management, audit management, authority management, account management and security authentication functions. Meanwhile, AS and GS undertake the channels for user data transmission, such as DCH, for transmitting service data; the control plane integrates all the channels for control data transmission, system resources management, allocation, commands and status and other control data. The user plane and control plane guarantee the transmission of the data path between the GS and the GSC, providing the end-to-end data transmission of ABCS. The communication process between the SGW and the GSC undertakes the associated functions such as authentication, authorization, account management, and key management of data, entities, and users, which provides a comprehensive guarantee of the secure communication between air and land and the management of protocols; the air-port transmission between the GS and the AS The air-port transmission between GS and AS is based on the channels in the FL and RL in the physical layer (PHY), realizing the air-ground wireless communication with high dynamic movement. For relevant security functions that need to reside in different entities, the security functions mainly rely on Sub-Network Protocol layer (SNP) and Link Management Entity (LME). The on-board application security layer is responsible for completing the authentication and key negotiation between the on-board side and the security gateway, and is able to derive the negotiated session master key, derive the encryption key and integrity protection key used to encrypt and protect the integrity of the user data, as well as to send the derived control data protection key to the on-board LME security layer. The ground application security layer and the on-board application security layer have similar functions, and are responsible for identity authentication and key negotiation with the security gateway, and derive the key used to protect the user data and control data to ensure the safe transmission of data. Security functions for user data are integrated in SNP, such as encryption of user data payload or protection of user data integrity and authenticity. Control data security is realized in LME and Medium Access Layer.

The ABCS SNP complements the higher layer approach as it creates the ability to protect the data chain control data, which in contrast to user data, is only exchanged between ABCS devices and is not forwarded to higher layers. Therefore, higher-level security methods are unlikely to help protect this control data.

B. User Data Encryption and Integrity Protection Scheme for ABCS

The security function of ABCS is provided by the SNP. The data from the upper layer can be encrypted directly in this layer without relying on the instructions of the lower layer protocol. This arrangement helps to better utilize the advantages of end-to-end secure transmission protocols. Therefore, when designing the security data message format of the ABCS, the requirements of the subnet layer protocol should be followed. In order to ensure the transmission security of ABCS, the security protocol needs to add fields of confidentiality, integrity and source authentication protection to the message structure. Table 1 shows each field contained in the SNP layer data message. Each SNP_PDU should include the security header part, data part and end part to meet the confidentiality and integrity protection of data. For MAC fields, different byte sizes can be set according to different user needs, such as 128bits or 256bits. The fields of the security data message format of ABCS are shown in Table 1.

TABLE I. SECURITY DATA MESSAGE STRUCTURE

Field name	Field length	Field description
CTRL	1 bit	Data type
SEC	3 bits	Message authentication code length
SEC_Mode	2bits	Security protection mode
NSEL	4 bits	Network Selector
Data	Variable	User data payload
SQN	24 bits	Serial number
MAC	0/128/256 bits	Message authentication code

SEC_Mode is used to represent the data types of different security services, as shown in Table 2.

TABLE II. SEC_MODE FIELD FORMAT

Field name	Field value	Field description
SEC_Mode	00	Encryption
	01	Integrity protection
	10	Encryption and integrity protection
	11	Message source authentication

According to the data integrity protection requirements of ABCS, this paper uses SM4 algorithm to encrypt and decrypt, and uses domestic hash cipher SM3 algorithm to realize the integrity check of communication data, so as to prevent data from being tampered with, thus posing a security threat to

aviation operation. According to the characteristics of user data transmission, this study combines the data security message format proposed in the previous study to design the process of data integrity protection and verification, as shown in Fig. 4.

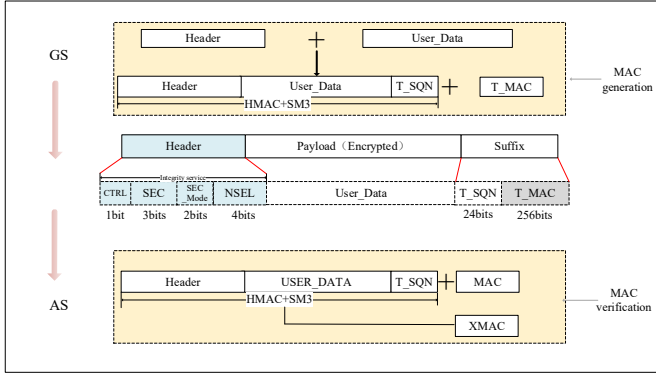


Fig. 4. User data security protection process

Fig. 4 shows the process of integrity protection and verification for user data message format. The header, Message Authentication Code (MAC) and Protocol Data Unit (PDU) of user data are encapsulated and transmitted respectively. The entity sender configures the corresponding integrity algorithm according to the security policy of header to generate MAC. The GS entity has judged the value of the Ctrl field during the encryption process, followed the principle of first encryption and then integrity protection, and called the integrity interface. That is, the integrity protection and integrity MAC value of the given data message are verified. The field value of the complete algorithm suite is obtained, and it is assumed that the HMAC-SM3-256 algorithm is used to generate the MAC value. Then the encrypted ciphertext and MAC are sent to the aircraft side. The aircraft receives the data packet to calculate whether the XMAC matches the MAC. If the matching is consistent, it is proved that the message has not been tampered with during transmission.

C. User Data Message Source Authentication Protection Scheme for ABCS

In ABCS, the identity authentication of data sources is very important, because these data are vulnerable to the threat of replay attacks by unauthorized illegal users, and face the risk of data forgery during transmission. The simple way to verify broadcast packets is to attach a digital signature to each packet, which will require the use of public and private key methods to verify. This leads to two disadvantages: First, the operation of generating and verifying digital signatures is several orders of magnitude higher in computational cost than performing symmetric operations (such as generating message authentication codes). Secondly, the size of the digital signature is currently about 512 bits, which will generate significant overhead. If MAC is used, the security overhead can be reduced. Therefore, considering the use of symmetric keys to achieve asymmetric authentication can greatly improve the computational and communication traffic, and avoid the problem of packet loss in asymmetric methods. The TESLA source authentication scheme overcomes the symmetric weakness of MAC through key delay. Therefore, this paper proposes to implement the TESLA source authentication scheme in combination with the national secret algorithm SM series. To ensure the data source authentication

when transmitting user data to the SNP, the TESLA protocol is used to authenticate the communication source between GS and AS. Its deployment in ABCS has the following key points: ensuring loose time synchronization between AS and GS; after GS receives the user data sent by AS, it must be buffered for a period of time before receiving the authentication key for authentication. The implementation principle of ABCS TESLA is shown in Fig.5.

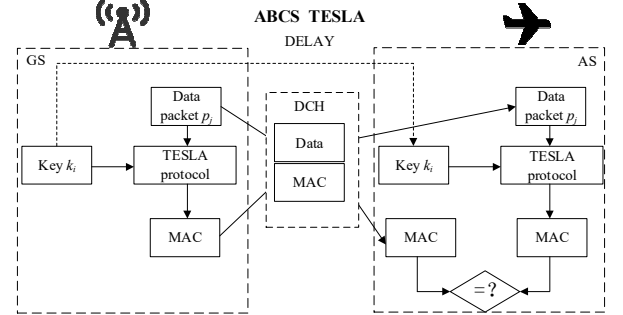


Fig. 5. The implementation principle of ABCS TESLA

IV. ANALYSIS OF ABCS SECURITY PROTECTION SCHEME

This section uses the aviation broadband communication security communication test platform to test the function of the aviation broadband communication system security protection mechanism, and simulates the attack end to test the security of the aviation broadband communication system in the data transmission process to verify whether it meets the security requirements.

A. Test Environment

This scheme uses a hardware such as an industrial computer and two servers. The server simulates GS and AS respectively. The industrial computer is a security gateway. According to the design of the above software security module, this test platform is mounted on the server to establish a communication connection. The specific deployment of the physical connection of the test equipment is shown in Fig. 6.

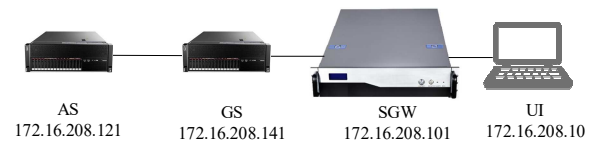


Fig. 6. Physical connection diagram of test environment

B. Performance Analysis

In this paper, the SM4 algorithm is used to guarantee communication data confidentiality. Since the aviation communication message is not a fixed length, it consists of 75% small packet data (about 270 bytes on average) and 25% large packet data (about 1400 bytes) [15]. In order to verify the performance of the proposed algorithm, the SM4-CBC, SM4-GCM, AES-HMAC-SHA256 algorithm of the password card is called for testing and analysis. The test results are displayed in the Fig. 7, Fig. 8. The results show that the average encryption/decryption time required for data of different plaintext lengths in the SM4-CBC mode is the shortest. The average time required for encryption is 190.888μs, and the average time required for encryption is

190.954μs, which meets the requirements of low latency and high speed for ABCS. Therefore, SM4-CBC encryption mode is used to encrypt and protect the data of ABCS to ensure the confidentiality of communication data.

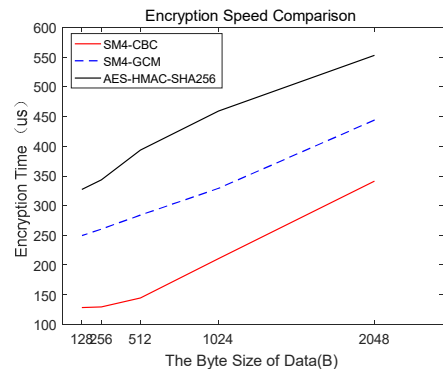


Fig. 7. Encryption Speed Comparison

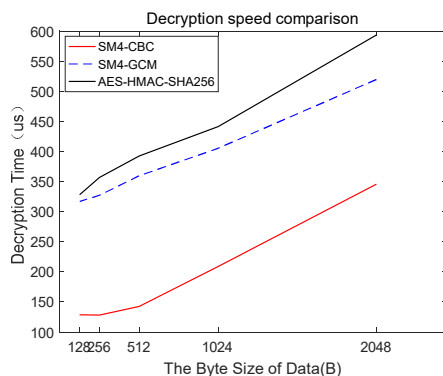


Fig. 8. Decryption speed comparison

In order to verify that the functionality and performance of the proposed algorithm meets the requirements, the SM3-HMAC algorithm of the cipher card is invoked for comparative analysis. In the experiment, different bytes of data packets are used to generate two message authentication codes of different sizes, namely 128 bits and 256 bits. The experimental results are displayed in Fig. 9, which shows that the time required to generate the message authentication code using SM3-HMAC-256 is almost the same as the time required to use SM3-HMAC-128. Therefore, using this algorithm to realize the function of integrity verification can meet the needs of ABCS.

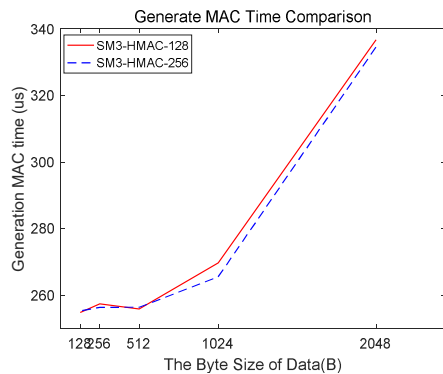


Fig. 9. Decryption speed comparison

C. Compliance Verification

Based on the protocol stack architecture, a protocol stack test platform for testing security protocols is developed. The

ABCS test platform includes: AS, GS, SGW. The protocol stack architecture currently designed and implemented is shown in Fig. 10.

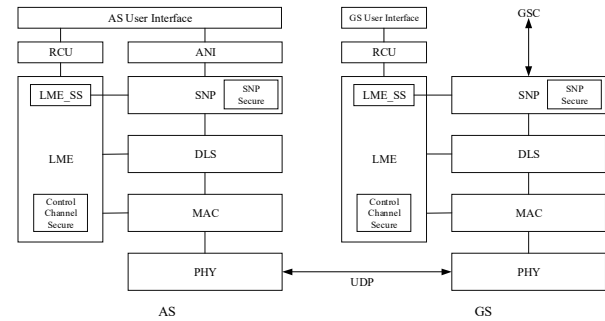


Fig. 10. The protocol stack architecture of ABCS

This system protocol stack and uses UDP broadcasts to simulate physical layer signal propagation. The core modules of this testbed are LME security module and SNP security module. First, GS and SGW establish a persistent connection, after which it broadcasts its presence to all AS entities in the LAN, the AS sends the first packet to GS and establishes a logical UDP channel based on the content carried by this UDP connection, and the GS acquires the packet and forwards it according to the upstream and downstream directions. When the GSC acquires the downlink packet, it parses and organizes the key information of the packet into a JSON format string to be sent to the security gateway; the instructions issued by the security gateway are first sent to the GSC to form a security authentication packet, which is then routed through the AC/R route to find the corresponding GS, and then transmitted upwards to the AS through the ground station, thus completing the single data flow process.

The protocol stack simulation test platform is used to test the compliance of the load security encapsulation protocol. After the AS and GS instances are started, the Wireshark is used to capture the packets. The result of the capture is the integrity protection of the DCH message to obtain 256 bits of MAC. The capture field corresponds to the Fig. 11, which conforms to the design of the user data security protection scheme.

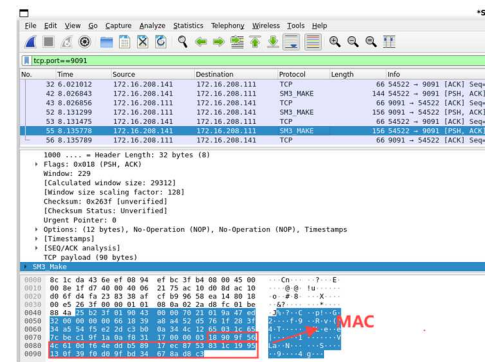


Fig. 11. DCH data example diagram

D. Safety Analysis

Confidentiality and integrity: The encryption key is used to encrypt the message, and by using the SM4 symmetric algorithm, the confidentiality of the message is guaranteed in the transmission process, preventing eavesdropping by intermediaries; the integrity protection key is used for the integrity protection of the message and checksums, and the

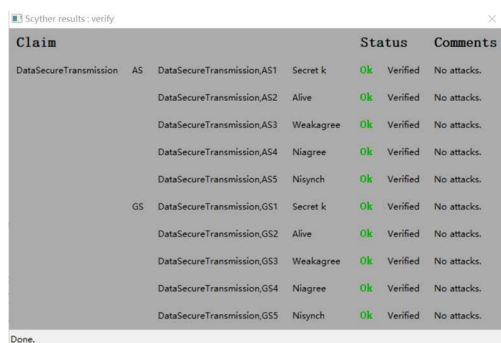
SM3 algorithm is used to compute the message authentication code MAC, and checksums compare the value of the MAC to determine whether the Whether the message has been tampered with or not.

Man-in-the-middle attack and anti-replay attack: the SQN field in the user data transmission message has an adder function, and the field SQN is updated every time the SNP message is sent, and the receiver determines whether it encounters a replay attack and man-in-the-middle attack by comparing the size of the SQN.

In this paper, Scyther formal verification tool is used to provide mathematical proof level guarantee for the security of communication protocol in aviation broadband security transmission technology by modeling protocol logic and automatically analyzing potential attack paths. Scyther is based on Dolev-Yao attacker model and model checking technology, which is used to automatically analyze protocol logic vulnerabilities (such as man-in-the-middle attack, key leakage, etc.) and generate attack path graphs (including role interaction, attacker tampering behavior).

Scyther defines the protocol flow through the role modeling language. 'Claim' represents the declared security attributes, mainly including the following: 'Secret' is used to verify confidentiality, which can determine whether the confidentiality of data is destroyed, that is, confidentiality; 'Alive' is used to verify that the protocol participants do perform the protocol steps, that is, authentication; 'Weakagree' is used to determine whether the two parties agree on the parameters of the agreement, that is, consistency.

In this paper, the security analysis of the data security protection scheme designed above is carried out, and the formal verification results of Scyther software are shown in Fig. 11.



Claim	Status	Comments
DataSecureTransmission AS Secret k	Ok Verified	No attacks.
DataSecureTransmission AS2 Alive	Ok Verified	No attacks.
DataSecureTransmission AS3 Weakagree	Ok Verified	No attacks.
DataSecureTransmission AS4 Niagree	Ok Verified	No attacks.
DataSecureTransmission AS5 Nisynch	Ok Verified	No attacks.
GS DataSecureTransmission GS1 Secret k	Ok Verified	No attacks.
DataSecureTransmission GS2 Alive	Ok Verified	No attacks.
DataSecureTransmission GS3 Weakagree	Ok Verified	No attacks.
DataSecureTransmission GS4 Niagree	Ok Verified	No attacks.
DataSecureTransmission GS5 Nisynch	Ok Verified	No attacks.

Fig. 12. Scyther formal verification results

The Scyther verification result interface shows that the tested data security transmission protocol satisfies the formal model: all declared (Secret, Alive, Nisynch, etc.) states are 'Ok', no exploitable vulnerabilities are found, and the protocol design is complete. It is suitable for high-dynamic and high-security aviation communication scenarios and provides theoretical guarantee for actual deployment.

V. CONCLUSION

As aviation mobile communication technology is gradually replacing the application of aviation narrowband data links, the research of ABCS has become crucial. Aviation data link information is transmitted on an open channel and is vulnerable to network attacks. To this end, this paper adopts

the national secret SM series algorithm, and puts forward the user data and control data security protection scheme of ABCS. In the protection scheme of ABCS, the average time overhead of confidentiality does not exceed 190.89 μ s, and the average time overhead of integrity does not exceed 273.60 μ s. The protocol's viability and effectiveness against tampering attacks are demonstrated by the experimental findings. In the future work, we will further optimize the security protection scheme of user data and control data, and minimize the security overhead under the premise of security assurance.

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REFERENCES

- [1] EUROCONTROL, "EUROCONTROL's Challenges of Growth 2008 Study Report", 2008.
- [2] SESAR, "European ATM Master Plan," SESAR Joint Undertaking, Edition 2, Oct. 2012; <https://www.atm-masterplan.eu>.
- [3] JAMES M. HALL E. Aeronautical mobile airport communications system (AeroMACS) : NASA/TM-2011-217236 [R] .[S. l.] : NASA , 2011.
- [4] Jackson, Michael R. C. et al. "4D Trajectory Based Operations — Speed Control Interoperability." 2019 IEEE/AIAA 38th Digital Avionics Systems Conference (DASC) (2019): 1-7.
- [5] China Civil Aviation Administration. Implementation roadmap of CAAMS : IB-ATMB-2020-001 [R]. Beijing : China Civil Aviation Administration, 2020.
- [6] B. Mahmoud, A. Pirovano, and N. Larrieu, "Aeronautical communication transition from analog to digital data: A network security survey," *Comput. Sci. Rev.*, vol. 11, pp. 1–29, May 2014.
- [7] Mäurer, Nils and Arne Bilzhaue. "Paving the way for an it security architecture for LDACS: A datalink security threat and risk analysis." 2018 Integrated Communications, Navigation, Surveillance Conference (ICNS) (2018): 1A2-1-1A2-11.
- [8] X. Li, Q. Zhang, L. Xu and T. Shang, "A Compatible and Identity Privacy-preserving Security Protocol for ACARS," 2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), Wuhan, China, 2022, pp. 1048-1053.
- [9] N. Mäurer, T. Gräupl, C. Schmitt, G. D. Rodosek and H. Reiser, "Advancing the Security of LDACS," in *IEEE Transactions on Network and Service Management*, vol. 19, no. 4, pp. 5237-5251.
- [10] National Cryptographic Administration. SM4 Packet Cryptography Algorithm <https://std.samr.gov.cn/hb/search/stdHBDetailed?id=8B1827F22B99BB19E05397BE0A0AB44A>, (2012).
- [11] National Cryptographic Administration. SM3 hash algorithm https://www.oscca.gov.cn/sca/xxgk/201012/17/content_1002389.shtml, (2010).
- [12] A. Perrig, R. Canetti, J. D. Tygar, and D. Song, "The TESLA broadcast authentication protocol," *RSA Cryptobytes*, vol. 5, pp. 2–13, Summer/Fall 2002.
- [13] N. Mäurer, T. Gräupl and C. Schmitt, "Efficient Control-Channel Security for the Aeronautical Communications System LDACS," 2023 IEEE 24th International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM), Boston, MA, USA, 2023, pp. 407-412.
- [14] U. Epple and M. Schnell, "Overview of legacy systems in L-band and its influence on the future aeronautical communication system LDACS1," in *IEEE Aerospace and Electronic Systems Magazine*, vol. 29, no. 2, pp. 31-37.
- [15] T. Gräupl and N. Mäurer, "An Air Traffic Management Data Traffic Pattern for Aeronautical Communication System Evaluations," 2019 IEEE/AIAA 38th Digital Avionics Systems Conference (DASC), San Diego, CA, USA, 2019, pp. 1-6.
- [16] Cremers, C. *The Scyther Tool: Verification, Falsification, and Analysis of Security Protocols - Tool Paper*. Springer, 2008, pp. 414–18.